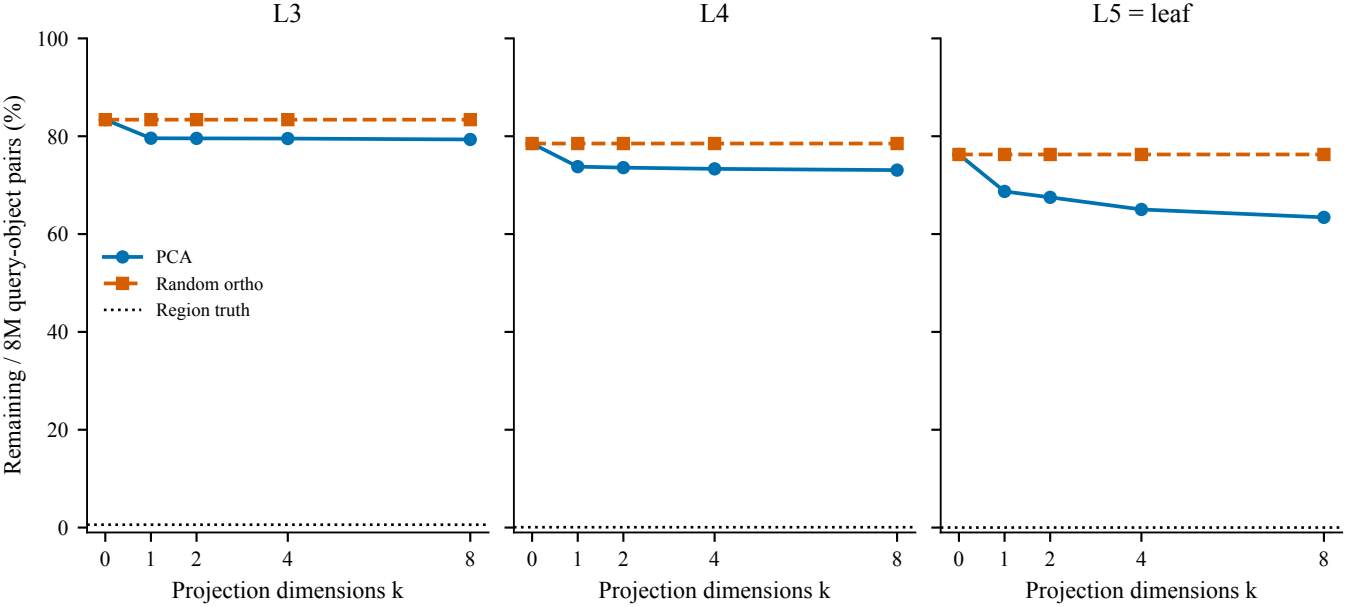
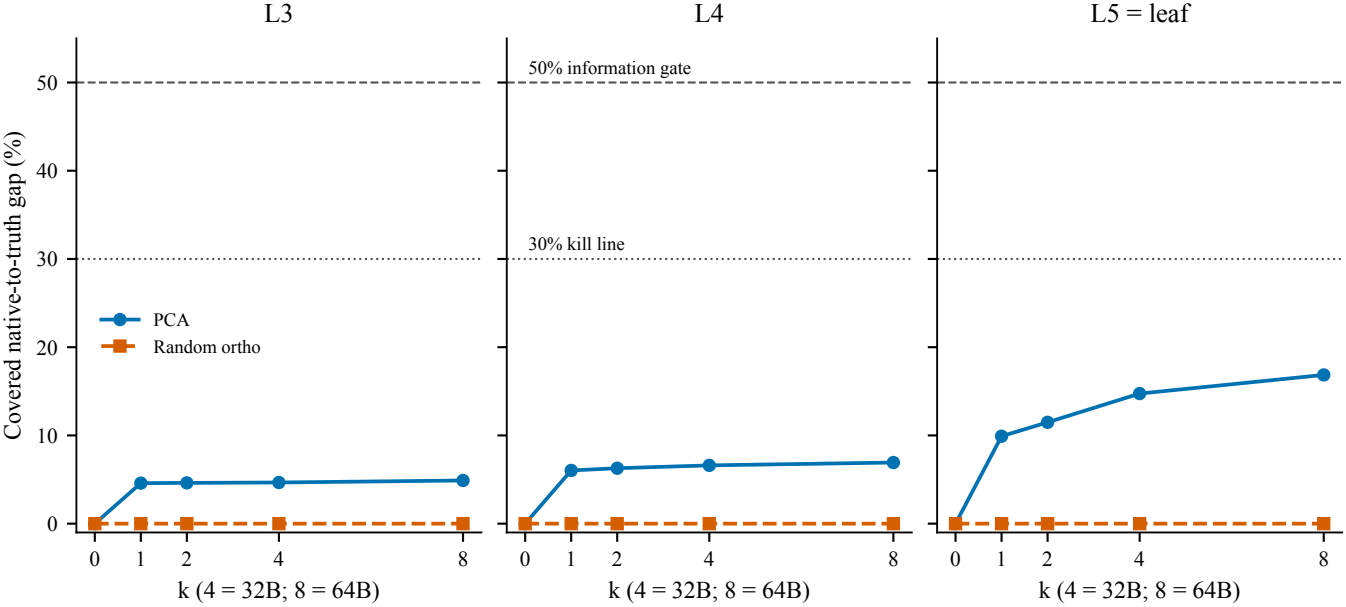
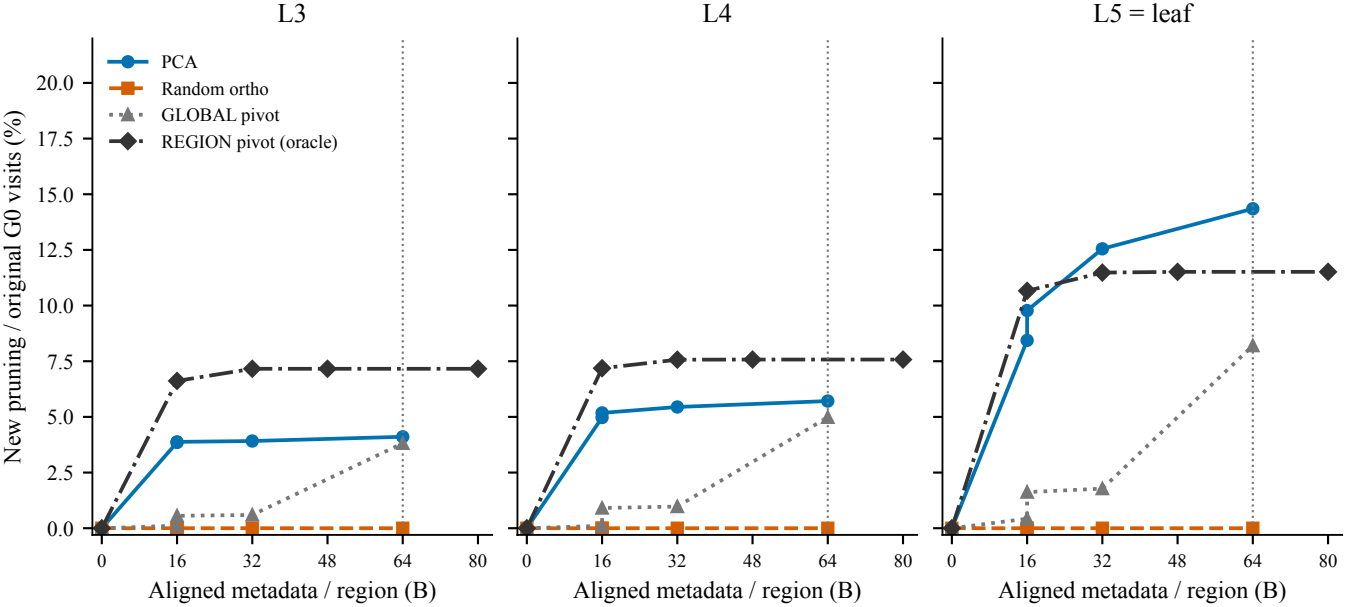
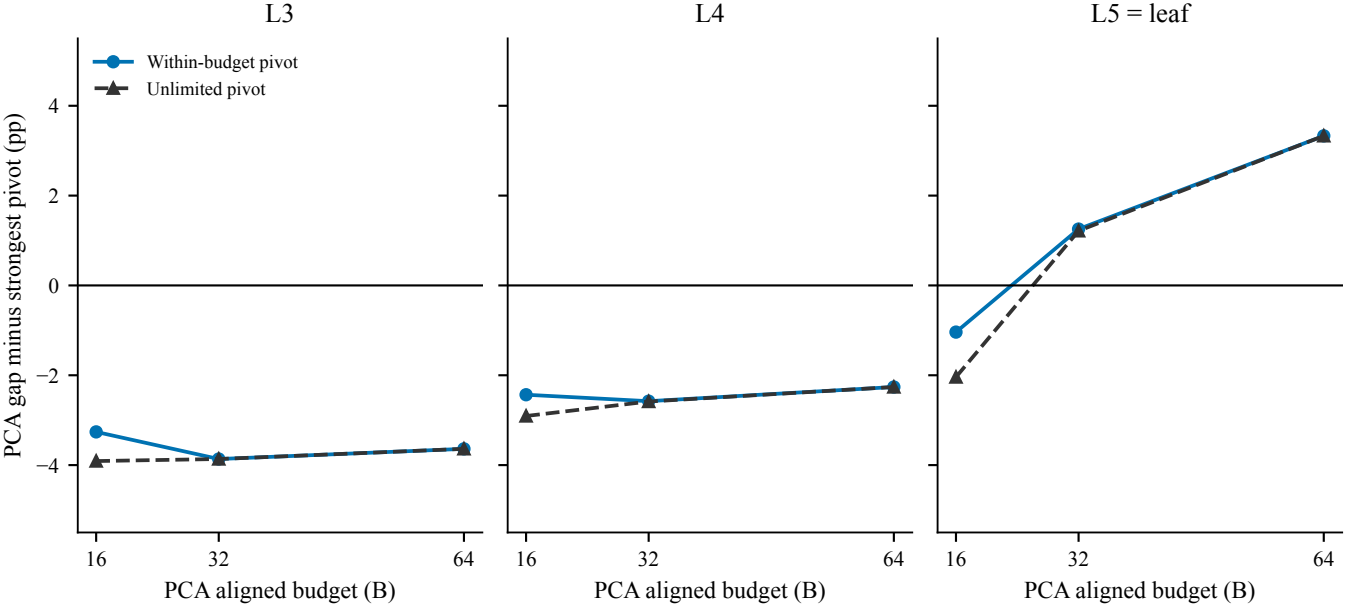

L3
L4
L5 = leaf
Remaining / 8M query-object pairs (%)
100
80
60
40
20
0
PCA
Random ortho
Region truth
0
1
2
4
8
Projection dimensions k


L3
L4
L5 = leaf
50% information gate
30% kill line
PCA
Random ortho
Covered native-to-truth gap (%)
0
10
20
30
40
50
0
1
2
4
8
k (4 = 32B; 8 = 64B)


L3
L4
L5 = leaf
PCA
Random ortho
GLOBAL pivot
REGION pivot (oracle)
New pruning / original G0 visits (%)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
16
32
48
64
80
Aligned metadata / region (B)


L3
L4
L5 = leaf
Within-budget pivot
Unlimited pivot
PCA gap minus strongest pivot (pp)
4
2
0
−2
−4
16
32
64
PCA aligned budget (B)